

Contact

amorillo@utb.edu.co

www.linkedin.com/in/anderson-morillo-792515153 (LinkedIn)

Top Skills

Amazon Web Services (AWS)

AI AWS

AWS

Languages

English (Full Professional)

Spanish (Native or Bilingual)

Certifications

AWS AI Practitioner Challenge

Marco web Django

Publications

VerbaNexAI Lab at SemEval-2024
Task 1: A Multilayer Artificial
Intelligence Model for Semantic
Relationship Detection

VerbaNexAI at SemEval-2025
Task 3: Fact Retrieval with Google
Snippets for LLM Context Filtering to
identify Hallucinations

PubMed API and LLM-Driven Hybrid
Retrieval System for Biomedical
Question Answering

UTB-NLP at SemEval-2023 Task
3: Weirdness, Lexical Features for
Detecting Categorical Framings, and
Persuasion in Online News

Anderson Morillo

AI engineer | NLP Researcher

Cartagena, Bolívar, Colombia

Summary

I build NLP and AI systems end to end, from RAG-based chatbots and fine-tuned transformers to real-time data platforms. My work spans LLM-powered conversational AI and HuggingFace model development, and production deployment with Docker, FastAPI, and Azure. I've shipped a meteorological monitoring system on TimescaleDB, engineered ETL pipelines in Databricks and PySpark, and published NLP research at three SemEval workshops through the VerbaNex AI Lab. I focus on shipping software that works in production.

Experience

COTECMAR

AI Engineer

April 2026 - Present (4 months)

Colombia

- Leading scalable deployment of a virtual assistant with user-trained local LLMs, combining on-premise fine-tuning and RAG for accurate responses on specialized technical documentation.

- Designed a separate work order management system with operational KPI tracking, automating task assignment and real-time performance metric monitoring.

Universidad Tecnológica de Bolívar

4 years 1 month

Research Assistant

July 2024 - Present (2 years 1 month)

- Engineered evaluation pipelines for hallucination detection in LLMs, implementing automated fact-retrieval and context-filtering systems.

- Fine-tuned transformer models (BERT, DeBERTa) for semantic framing and persuasion detection, optimizing inference latency via ONNX export and quantization.

- Built reproducible experimentation frameworks in Python (PyTorch, HuggingFace, scikit-learn) supporting dataset curation, bootstrap statistical testing, and manuscript preparation.

Student Research Assistant

July 2022 - June 2024 (2 years)

Cartagena, Bolívar, Colombia

- Developed multilayer NLP architectures for semantic relationship detection and lexical framing analysis, training and evaluating models on benchmark news corpora.

- Built end-to-end data processing pipelines (ETL, feature engineering, tokenization) using Python, Pandas, and Jupyter; reduced data preprocessing time by automating annotation workflows.

Freelance

AI Architecture Consultant

January 2026 - January 2026 (1 month)

- Designed the AI architecture for a multi-city meteorological-hydraulic monitoring platform (Montería, Cartagena, Medellín).

- Proposed data processing pipelines for real-time ingestion, cleaning, and analysis of hydrometeorological variables.

- Structured an early-warning alert system based on predictive models and automated thresholds.

COTECMAR

AI Engineer

September 2025 - December 2025 (4 months)

Colombia

- Architected a domain-specific conversational AI system using LLMs and retrieval-augmented generation (RAG); grounded responses in internal technical documentation to reduce hallucinations.

- Implemented chatbot backend services in Python (FastAPI) and integrated workflow automation via n8n, enabling real-time task orchestration and user interaction tracking.

- Engineered prompt strategies and context-window management to align model outputs with maritime/defense domain knowledge and user intent.

Del Laboratorio al Campo

Software Engineer

July 2024 - August 2025 (1 year 2 months)

Colombia

- Architected end-to-end meteorological data platform ingesting real-time sensor streams into TimescaleDB; served processed data via REST APIs to early-warning alert systems.
- Built interactive data visualization interfaces with Streamlit and optimized time-series queries for sub-second dashboard rendering across geospatial datasets.
- Designed ETL pipelines and database schemas for high-frequency IoT data; implemented automated alerting logic improving disaster-response notification speed.

ESENTTIA

IT infrastructure intern

September 2023 - March 2024 (7 months)

Colombia

- Built scalable ETL pipelines in Azure Data Factory and Databricks (PySpark), processing operational datasets for downstream predictive modeling and analytics.
- Optimized Spark SQL transformations and data partitioning strategies, improving pipeline runtime efficiency and ensuring schema consistency for ML feature stores.
- Supported Azure cloud infrastructure monitoring, security compliance, and data warehousing architecture; authored technical documentation for data lakehouse standards.

Education

Universidad Tecnológica de Bolívar

Master's degree, Artificial Intelligence · (August 2024 - June 2026)

Universidad Tecnológica de Bolívar

Systems and Computer Engineering · (August 2019 - March 2024)